



PEAK DEFENSE:
BUILDING ADAPTIVE SYSTEMS
FOR MODERN THREATS



PR3TACK

The Preemptive Tactics &
Countermeasures Knowledgebase

Bridging the anticipatory gap in
cybersecurity

Introduction

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PR3TACK

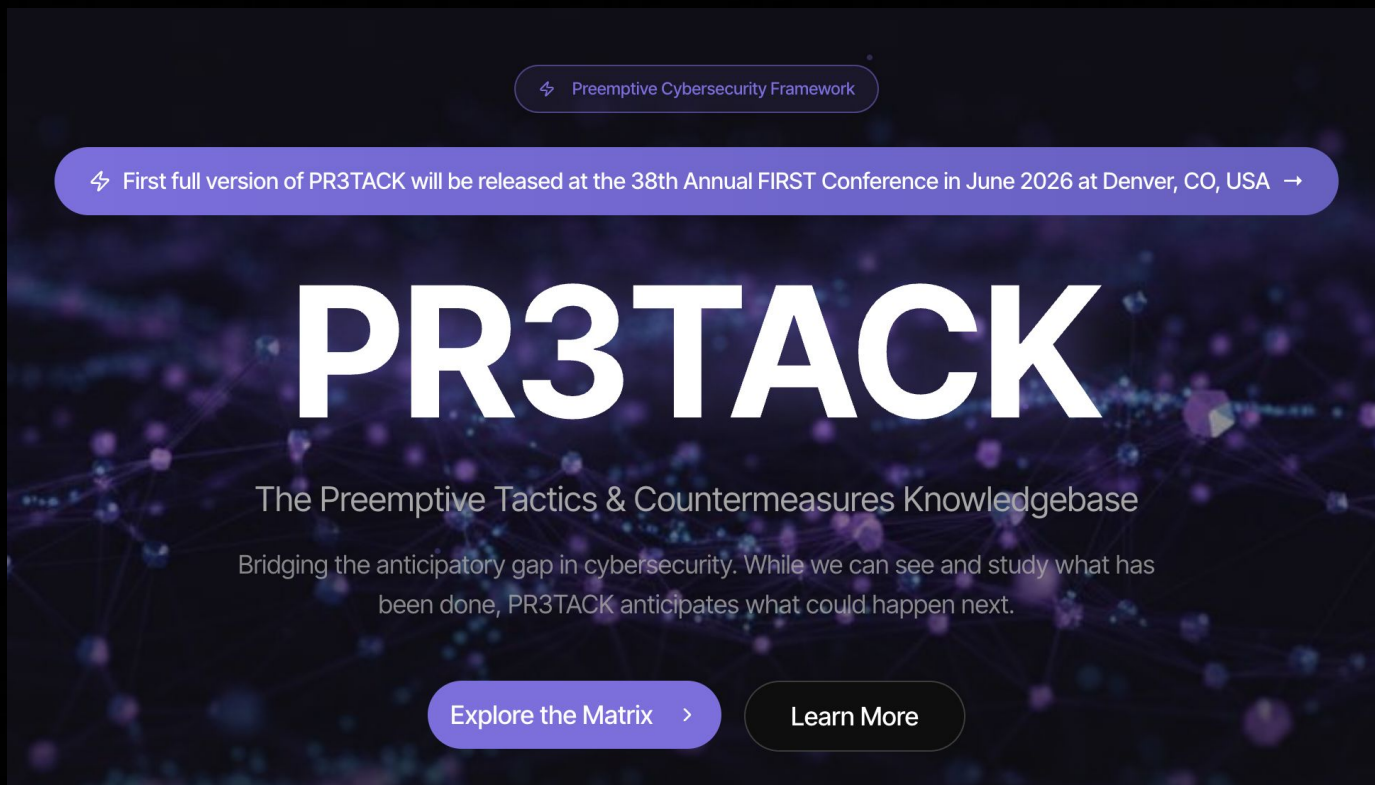
Preemptive Tactics & Countermeasures Knowledgebase

Preparing for Tomorrow's Attacks — Today

"A community-driven framework of plausible, technically feasible adversary techniques before they are observed in the wild."

pr3tack.org

What is PR3TACK?



⚡ Preemptive Cybersecurity Framework

⚡ First full version of PR3TACK will be released at the 38th Annual FIRST Conference in June 2026 at Denver, CO, USA →

PR3TACK

The Preemptive Tactics & Countermeasures Knowledgebase

Bridging the anticipatory gap in cybersecurity. While we can see and study what has been done, PR3TACK anticipates what could happen next.

Explore the Matrix >

Learn More

Why PR3TACK?

- **Reactive Fundamentalism:** Cybersecurity is fundamentally reactive — defenders learn from past breaches.
- **Documentation Lag:** MITRE ATT&CK mostly documents what has already been observed.
- **Persistent Latency:** Gap between adversary innovation and defensive adaptation leaves defenders in arrears.
- **Operational Edge:** Adversaries exploit this gap to maintain a strategic advantage.
- **Asymmetric Challenge:** Adversaries need one successful exploit; defenders must anticipate an unbounded threat set.
- **Absence of Foresight:** Plausible-but-unobserved techniques remain absent from frameworks until empirical confirmation.

Impact of Knowledge Gaps



Atom Table Abuse

Enabled stealthy execution in Windows environments.

Was widely exploitable but absent from ATT&CK, leaving defenders without a reference point.



Revix ESXi Ransomware

Technique: "Kill-all-VMs-before-encryption".

Dismantled quickly, so it never entered standard frameworks, creating a permanent blind spot.

What is PR3TACK?

- **Openly accessible framework** cataloguing plausible but unobserved adversary TTPs
- **Bridges the anticipatory gap** left by retrospective frameworks
- **Not speculation** — grounded in POC research, adversarial trends, and technical reasoning
- **Complements MITRE ATT&CK and D3FEND** with an anticipatory layer
- **Draws inspiration from foresight methodologies** in strategic studies, resilience engineering, and speculative design
- **Systematises a domain** that has been informal and fragmented until now

From Reactive to Preemptive Defence

The dominant defensive paradigm: breach occurs → analysis follows → mitigations deployed

This cycle codifies empirical fact but leaves defenders continually in arrears.



The Adjacent Possible

PR3TACK introduces a forward-looking dimension: codifying attack vectors that could emerge given current technological affordances.



Anticipatory Layer

PR3TACK does not replace reactive defence — it augments it with an anticipatory layer to proactively address future threats.

PR3TACK Ethical Policy

Plausibility, Not Speculation

PR3TACK is not a repository of science fiction. It relies on demonstrable plausibility, grounded in proof-of-concept research, adversarial innovation trends, and the technical properties of modern systems.

Priority Levels

High: TTPs demonstrated through proof-of-concept implementations

Medium: TTPs hypothesised with technical reasoning, awaiting validation

Low: TTPs emerging from academic research or ongoing investigation

Key Features

- Organised around tactics and techniques (overlapping with ATT&CK + novel categories)
- Each technique includes: description, feasibility assessment, and preemptive defences
- Seed Matrix v1.0 covers 17 tactic categories
- Interactive Matrix Navigator for exploration and custom JSON uploads

Ethical Policy

"If we cannot help defenders stop it, we will not help attackers learn it"



Mandatory Countermeasures

Every TTP submission must include viable countermeasures — no exceptions.



Strict Rejection Criteria

Submissions are rejected if countermeasures are missing, impractical, or ineffective.

Benefits of PR3TACK

Complementarity with Existing Frameworks

- **ATT&CK** catalogues what is known to have occurred
- **PR3TACK** catalogues what is plausible but not yet observed
- **D3FEND** specifies defensive measures against documented techniques
- **PR3TACK** extends this by suggesting preemptive countermeasures against as-yet-unrealised threats
- *Together, these frameworks create a layered epistemology of defence: empirical, reactive, and anticipatory*

PR3TACK Matrix Navigator

The PR3TACK Matrix provides an interactive way to explore the Seed Matrix — the "what could happen next" layer of threat modelling.

- **Visual Grid:** Column-based view with tactics and related techniques
- **Search & Filter:** Across tactics, techniques, platforms, and status
- **Interactive Details:** Click techniques for detection ideas and mitigations

- **Extensibility:** Upload custom JSON files to extend the Seed Matrix
- **Proactive Defense:** From a static reference to a living, interactive tool
- **Team Mapping:** Map findings for detections, IR, and research

PR3TACK Matrix Navigator

 PR3TACK Navigator					
COLOR SEARCH Find technique... FILTER All High Medium Low					
Execution	Persistence	Privilege Escalation	Defense Evasion	Command & Control	Discovery
T1.001 BIOS Config Trigger Payloads Low	T2.001 Abuse of Windows Atom Tables High	T3.001 Exploit Virtual Device Drivers (VDDs) Medium	T4.001 Cloud Metadata API Overloading Medium	T5.001 Bluetooth HID Covert Channels Low	T6.001 Browser Extension Enumeration via IPC Abuse Medium
T1.002 Peripheral Firmware Stager Medium	T2.002 Hidden Service Registration Medium	T3.002 Credential Store Forging Medium	T4.002 Telemetry Tampering Medium	T5.002 Covert DNS-over-HTTPS Tunnels Medium	T6.002 Identity Topology Mapping Medium

Technique Examples from the Seed Matrix

Tactic	Technique	Feasibility	Preemptive Defence
Execution	Peripheral Firmware Stager — benign-seeming firmware stages payloads	Medium	Firmware verification, device attestation, vendor allowlisting
Persistence	Abuse of Windows Atom Tables — store payloads for retrieval	High	Monitor GlobalAddAtom/GetAtomName APIs, EDR hooks, limit usage
Pre-Positioning	OSS Supply-Chain Sleeper — benign code with dormant triggers	Medium	Code review gating, provenance checks, CI/static analysis, vet dependencies
Cognitive Manipulation	Analyst Habituation Lures — high-volume alerts to normalise FPs	High	Adversarial training, alert hygiene, purple-team injects
Digital Exhaust	IOC/Telco Feed Pollution — mass submission of false indicators	High	Source reputation scoring, feed vetting, automated quality checks

Full Seed Matrix Overview

A comprehensive map of plausible next-generation adversarial techniques across 17 tactical categories.

Scope & Structure

- Maps 34 techniques across 17 tactic categories
- Standardized entries: description, feasibility, and preemptive defense
- Feasibility ratings: High, Medium, or Low

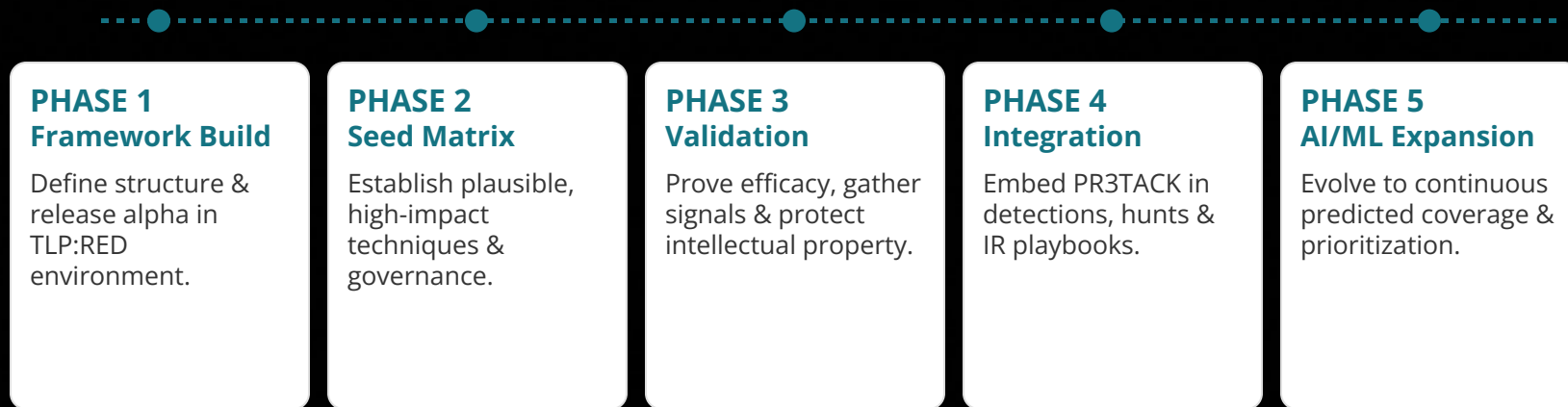
Technical & Strategic Breadth

- Hardware: BIOS config triggers & peripheral firmware
- Classic: Execution, persistence, and privilege escalation
- Strategic: Standards manipulation & investor narrative orchestration

Evolution & Novel Domains

- Covers novel areas: Governance subversion, cognitive manipulation, and AI/ML subversion
- Designed for extensibility: Community contributions drive continuous matrix expansion

Roadmap: The PR3TACK Evolution



Call to Action: Transitioning to Preemption

The history of cybersecurity has been one of reaction. While this remains necessary, it is insufficient in an era of accelerating adversarial innovation. **PR3TACK proposes a shift:** by cataloguing what could plausibly happen next, it enables defenders to move from reaction to preemption.

Organisational Impact

- Reduce risk, strengthen detection and IR.
- Avoid being tomorrow's case study.
- Proactive posture & enterprise resilience.
- Guide red/purple team exercises.

Industry Impact

- Pioneer predictive/preemptive cybersecurity.
- Influence global defensive posture.
- Common vocabulary for emergent threats.
- Guide vendor product development.

Culture & Academia

- Treat the "adjacent possible" as seriously as the observed.
- Close the latency gap and reduce adversarial advantage.
- Formalise the "plausible unknown" in CTI epistemology.

Together, we contribute to a more resilient digital ecosystem.

Ready to Get Started?

Join the community of security researchers anticipating the next generation of cyber threats. Contribute techniques, share insights, and stay ahead. Get your name on the PR3TACK researchers board!

This initiative is primarily for CSIRTs and other security teams in the industry. Currently we're only taking Expressions of Interest (EOIs) and will respond to requests once they are processed.

Contribute - Submitting PR3TACK TTPs

Community & Contribution

- **Collaborative project:** contributions invited from practitioners, academics, and organisations worldwide
- **Intake process:** requires contributor details, technique description, feasibility assessment, potential impact, and suggested preemptive defences
- **Supporting materials:** POC code and supporting research strongly encouraged
- **Recognition:** contributors may elect to be recognised publicly or remain anonymous
- **Curation:** Core Team curates submissions for quality and coherence

Get Involved with PR3TACK

Join our global network of practitioners and researchers to shape the future of preemptive defense.

Join the Team

Help us build the framework and expand our research reach.

join@pr3tack.org

Submit a TTP

Contribute new techniques to the Seed Matrix.

ttp@pr3tack.org

Submission Resources

Download the standardized submission template directly from our website.

Website: **pr3tack.org**

Thank you!

pr3tack.org

Vish and David

